

Engineering Activity Process
Rev B
May, 25th, 2012

A- Introduction

The EAP module will be used to manage the following process:

- Engineering Change Request / Engineering Change Notification for Product drawing correction : When engineering notification is issued without prior request from other department, it will be managed with an internal “engineering change request” (EAP opened by Engineering).
- “Minor project” development: minor adaptation for better produce-ability.
- Engineering information request, document change for typos or better legibility
- “Major project” development like New Product Development or complete upgrade of a TLD Product line, but also for Significant improvement/evolution/technologic change of a TLD Product line: **In that case MEAP will be used to track all the EAPs related to the project. See details in MEAP section below.** Specific EAPs will be opened for the phasing of the project.

Projects type 1 the validation is done at proposal level and notification level in the related EAP. Projects type 10, 100 & 1000 require critical validations at some point of time. For each phase a specific EAP will be open. These EAP will be used to propose the exit criteria for validation to the individual listed in the “TLD Group Engineering Guidelines”. These files will be posted in the Files section of the “phase” EAP (and not in the BP).

The EAP is a communication tool controlled within each factory by the Engineering Manager (EM). Like any TLD workflow module, the EAP system works at two levels:

- The EAP itself where description of the problem itself is done, pictures can be added as well as hyperlinks with P/N, WC, SB, NCR, CPA, ... These links are very important because they help to make information gathering more efficient when working on the EAP.
- The TASK level for each EAP. Everyone may be assigned a task and must proceed with his or her task and close it with comments when done.

In other words, some tasks can be opened and closed independently of the EAP itself.

B- Who can generate an EAP? :

- Any Factory department employee of a TLD factory can generate an EAP to its engineering department or to the Pilot Factory Engineering department for RANGER products.
- Any Spare parts department employee who is registered on the TLD Intranet, can generate an EAP to the engineering department of any TLD factory
- For new projects, the PSM and TLD management staff, in respect with the TLD rules (Minor or Major ones)
- Through a pertinent PDC that would end up generating an EAP (vector to be used by the Sales & Service organization that won't create EAP directly)
- By any approved TLD vendor after validation by the relevant TLD Material Manager
- Engineering department enter EAP to track proposed changed and make use of the workflow for proposal and notification.

Please note that whenever you enter an EAP, by default, you will be considered as the INITIATOR. In case you are entering a EAP on behalf of somebody else, please change the INITIATOR name to the originating person. If you do not, you will be sent ALL email notifications as the EAP progresses and will continue until it is closed.

C- How to submit an EAP:

EAP fields description:

- Factory: On RANGER product, when the RANGER base is affected by the request, the EAP should be assigned to the pilot factory. As some initiators may not affect it correctly, it is possible to change the factory when engineering manager changes status to "in progress".
- Category
 - Engineering Change,
 - Engineering development: (Minor or Major ones)
 - Engineering Information Request
- Importance factor :
 - 1 :supported improvement,
 - Sub MEAP
- If over night management is required, the EAP may be directed straight to the engineer in charge with the related equipment. This shortcut process is restricted for BOM update, qty modification, and easy modifications which doesn't require multiple tasks....
- Equipment type/mode
- Top bill number: Part number of the main component, assembly, sub-assembly affected with the modification, or BOM number. When multiple P/Ns are affected, please enter all P/N. This field will also be used for search purpose. If no number affected, use the product name.
- Short description: Please note that it is very important to put clear and precise information with pertinent wording in the EAP description to facilitate the research. **The EAP short description must be in English**
A prefix with equipment model name can be added when EAP is specific for this models to easier the research.
- Full Description: Can be done in local language. It shall describe the needs for modification, the origin(s) of the need, and can include a proposal to resolve and address the issue (the proposal cannot replace the description of the issue to address!).
- Files: EAP initiator may need to attach pictures, sketch, and modifications on drawings...
- Possible links with P/N, PDC, NCR, BOM, WC...etc

In case some information posted is unclear, the EM has the responsibility to clarify it and make the relevant changes, including the importance factor and affected P/N while turning it to IN PROGRESS.

D- EAP STATUS DEFINITION:

PENDING = When a request is just reported, the EAP status appears at PENDING.

If the request is judged pertinent by Engineering, the status will be changed to IN PROGRESS.

Please note that the EM should check pending EAP on a daily basis and change their status from PENDING immediately.

The overnight EAP appears as PENDING and will turn to NOTIFICATION status as soon as they are accepted by the engineer on charge with the equipment. It is so very important to make sure, before accepting the EAP, that the overnight status is legitimate and that the change requested doesn't need to be validated by MLM, or has no significant impact on cost. EM will be copied with the notification form.

IN PROGRESS = Engineering department is working on the request. At this stage, tasks can be assigned with due dates to personnel.

Depending on the magnitude of the EAP the task quantity and content may vary.

For simple modifications, there will be task related to engineering modification (at least one), possibly, a task for the cost impact (or estimation), and a task for the verification and document impact.

For more significant modifications, the tasks module is also use to assign tasks to other departments in the factory. Engineering can use it to get prototypes, to collect information on inventory status, and possible effectivity.

Once all the tasks are closed, the EAP can be turned to PROPOSAL status by the EM.

REJECTED = Request is rejected because not justified. Add a comment to explain to the Initiator why the request is rejected before rejecting it. Please make sure of not rejecting a EAP already taken into account in another EAP, please simply close it.

PROPOSAL= At this stage, the engineering activity is done with all the tasks closed. The EM will fill the field "Proposal Description" with a brief description of the change and the reason which drove it. He will also provide the estimated cost when it is significant.

The EM will then define the validation circuit and select in each department, PSM, MLM, PM, COO or misc, the one he wants to get involved in the validation process.

For RANGER products and for components used on other factory (ies) the circuit may include departments from other factory than the one from where the EAP is generated.

- RPD:

When the changes affect a RANGER product.

When the changes affect a critical component, a group standard component or system,...

- PSM :

When the propose change has an impact on the product aspect or performance for example.

When the change was originally requested through PDC or customer requests....

When the change will require a SB.

When the change has an impact on sales documentations, or price.

- MLM:

When the change has impact on current order, or inventory but also to inform on effectivity of the change.

To easier the typing, some proposal are pre-filled in the message block. The EM just needs to erase the non relevant sentences and to complete the suggested disposition actions (other info can be added):

- Effective from /à appliquer à partir de l'OF ###

- Following depletion of existing stock / à appliquer après épuisement des stocks

- Following depletion of existing stock and orders in process / à appliquer après épuisement des stocks et des commandes en cours

- Immediate by modification of existing stock & orders / stocks et commandes en cours à modifier

IF ACCEPTED, PLEASE INDICATE AS COMMENT THE EFFECTIVE DISPOSITION PLAN YOU INTEND TO PUT IN PLACE

- PM :

When the change has impact on "in process" production, or when it require new assembly procedures.

- COO:

When the changes have significant financial impacts (cost or saving).

To easier the typing, some proposal are pre-filled in the message block. The EM just needs to erase the non relevant sentences and to complete the suggested disposition actions (other info can be added):

Proposed Cost impact is above the current limit in the factory

or

Proposed Modification require Compulsory SB

- Other:

The EM can select anybody within TLD organization who needs to be included in the validation process.

At this stage, the process goes through a validation sequence where all the department in the validation process need to accept or reject it. Comment can also be added for clarifications.

If all the parties accept the proposal and provide additional information where required (disposition action from MLM, actual cost,...), the EM can turn it to NOTIFICATION.

If one or more of the parties involve in the validation process reject the proposal, it needs to be documented through clear comments, the EAP is returned to IN PROGRESS status.

NOTIFICATION= The EM will fill the field "Notification Description" with a brief description of the change and the reason which drove it. He will attach the related documents in the File block (Drawings, Doc, SB, manual,...).

When the cost impact is significant he will fill the designated block with currency and cost (+ or -)

The EM will then define the notification circuit and select in each department, PSM, MLM, PM, COO,SPM,CSM or misc, the one he wants to notify the change.

For RANGER products and for components used on other factory(ies) the circuit may include departments from other factory than the one from where the EAP is generated.

- PSM
- MLM : To indicate the agreed disposition action.
- PM : To indicate the action to consider.
- COO: When the change have significant financial impacts (cost or saving).
- SPM : When needed as when the change has an impact on item subject to spare part business. It has to clearly indicate the required actions on ERP update (SPH should record the EAP number to document the ERP modification), disposition actions: Following depletion of existing stock, Following depletion of existing stock and orders in process, Immediate by modification of existing stock & orders. The instructions have to be directive and may indicate that some items need to be maintain in inventory.

- CSM: When an SB is required or when the EAP is related to a significant change in a program/system with new sequences, or with impact on operations and maintenance.

- Other: The EM can select anybody within TLD organization who needs to be included in the notification.

At this stage, the process goes through a notification sequence where all the departments in the circuit need to acknowledge.

CLOSED = Change is now done and notified.

The EAP turn to close automatically and immediately once notified, and the system open individual tasks for acknowledgements.

Any changed status is recorded with the reason for this change. The reason recorded needs to be very clear and explicit, especially in the case of a closure. By only reading the Description and the Closing explanation, everyone should understand what's been done.

TASKS :

If multiple actions are required, even from one person, it shall be broken down in several independent tasks to better manage the follow-up.

Any task assignee can only transfer tasks to his subordinates.

Rescheduling a task must be supported by a very good reason. Personal workload or being late is not a good reason for a rescheduling.

TASK COLOR SIGNIFICATION:

GREEN = task in progress, still on time

RED = task is late, past due date

NO COLOR = Task completed and closed

COMMUNICATION AND SYSTEMATIC FEED BACK RULES:

The EAP module is set-up to provide systematic feedback to the initiator of the EAP and to the task assignees.

Therefore, the initiator is formally informed on how his/her EAP is progressing, what solutions have been found and what actions have been taken.

Automatic emails are sent according to the following rules:

Email to the initiator anytime a:

- Action/task has been posted for this EAP

- Comment has been posted for a EAP task

- Action status relative to this EAP has been modified

- EAP status itself has been modified

Email to action assignee anytime a:

- New task as been assigned to that person

7 days before an action is due
The day it was supposed to be due
Every 7 days after when the action is overdue
A comment has been added

In case of any overdue tasks, a warning message will also be sent to the manager of the assignee.

After 60 days of being overdue, any task is automatically escalated to the manager of the assignee.

Through this mechanism, ultimately a very much overdue task gets to the TLD group CEO.

CAUTION: Some EAP top users have access to the EDIT function. The use of the EDIT function does not create any email follow-up because EDIT actions are not recorded.

COMPANY NUMBER:

Company Number Factory

400 TLD WIN (Windsor)
420 TLD SHE (Sherbrooke)
500 TLD MTL (Montlouis)
520 TLD STL (St Lin)
600 TLD HKG (Hong Kong)
620 TLD TW (Taiwan)
640 TLD SHA (Shanghai)
660 TLD WUX (Wuxi)